

Target Tracking in Sensor Networks Using Energy Efficient Prediction Based Clustering Algorithm

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Abstract

In sensor network applications, energy is considered to be the most critical constraint. Our aim is to reduce the energy consumption as well as gain efficient accuracy by integrating the concept of tracking algorithm with heuristic approach. An energy efficient prediction-based clustering algorithm is being proposed here, based on the fact that the movements of the tracked objects are sometimes predictable. The proposed algorithm helps us to reduce the transmission distance between transmitter and receiver nodes and decreases the number of transmitted packets. Only the special nodes are being activated and rest of the nodes goes to sleep mode for energy saving. The proposed method has better performance than other prediction based methods in energy efficiency when simulated.

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Keywords: Tracking algorithm; Sensor network; Prediction- Based; Clustering algorithm; Energy efficiency.

1. Introduction

Wireless sensor network (WSN) is a network, which consists of a group of tiny sensor devices called *motest*; together with specific application software tailored for particular purpose and embedded software like Tinyos. Data collection is facilitated in sensor network, in an environment which is hazardous and area which is difficult to be discovered as they are included for specific job, therefore at a time only one application is made to be run on one sensor node. Tracking mainly includes monitoring and detecting the locations of Real-world objects, with the help of several types of sensing technologies such as acoustic, seismic, electromagnetic, etc and information is being sent to the base station for later analysis.

Many challenges have been faced in WSNs in our day today life. One of the main challenges is Energy efficiency. Since the Sensor nodes have limited battery, it's highly impossible to replenish their battery. Therefore the lifetime of a sensor node is strongly dependent on its battery lifetime, and these sensor nodes play a vital role in the lifetime of a WSN.

An energy efficient prediction based clustering algorithm for target tracking in WSNs is being proposed here. Target localization takes place locally in sensor nodes and not centrally in Cluster Heads (CHs). When localization is done locally, the selected sensor nodes calculate the current target location locally and one of them send this calculated location to CH, they do not send their sensed information from target to CH separately. Due to this the transmission distance between the transmitter and receiver is reduced in sending localization packets.

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The rest of this paper is organised as follows: In section 2 the previously existed tracking algorithms is made a survey. Section 3 carries the proposed algorithm. Then in section 4 we carry out the simulation and performance evaluation, and in section 5 the paper is concluded.

2. Related works

In recent years many target tracking algorithms are proposed. These algorithms are dependent on three main categories including tree-based, cluster-based and prediction-based. In tree-based methods the organization of the network is in the form of hierarchy tree. The network is divided into clusters to support collaborative data processing in Cluster-based method. In Prediction-based methods, a prediction mechanism is used to predict the next location of moving target. Some of the previously existing objects tracking algorithms are discussed below.

In [3] is presented a prediction based approach, called localized prediction. The localized prediction consists of a *localized sensor network architecture*, and a prediction mechanism called *dual prediction*. The network architecture consists of majority of sleep nodes and they are waken up by an active sensor node, via a low power paging channel, to follow the task of object tracking. Calculations are made about the predictions about future movement of a tracked object, at both of a sensor node and its cluster head. If the object's movement is consistent with the prediction, information collected at a sensor node is not being sent.

Authors in [2] proposed Prediction based Energy Saving Schemes (PES). PES includes three parts: 1) a prediction model which helps in the activation of sensor nodes that are expected to discover the object. 2) a wake up mechanism, that sets up which nodes and when they should be activated, based on some heuristics taking both energy and performance into account; 3) Finally when the network loses the track of an object a recovery mechanism is being initiated. The basic idea of PES is to make the sensor node stay in sleeping mode as long as possible, if it does not perform the duty of object tracking (i.e., there is no object in its detection area).

Algorithms that proposed in [2] are Scheduled Monitoring (SM) and Continuous monitoring (CM). In the scheduled monitoring (SM) technique, all the sensor nodes in the network are allowed to stay in sleep mode; only for a brief period of time, they change their status to active mode where their monitored area is being sensed and their findings are reported to the base station, provided both the sensor nodes and the base station are well synchronized. An Object Tracking Sensor network (OTSN) does not need to report its collected data all the time is the major concept introduced in this technique. All the sensor nodes, in this technique, are set to active mode for X ms, followed by a sleep mode period of $(T - X)$ ms, based on the application requirements. In other words, the sensor nodes become active only when they need to report to the base station. During the whole network operating time, this alternating behaviour of the network between sleep and active modes is set to take place. It is obviously clear that, a major savings in the consumed energy is accomplished with the implementation of this technique, compared with the naive technique which will be discussed below; this is because all the nodes in the OTSN network are scheduled to operate in sleep mode as long as possible and to stay in active mode for a very short time. The continuous monitoring (CM) technique differs from that of the schedule monitoring technique; where only one sensor node is made to continuously stay in an active mode, while detection and tracking of the object is done as long as this object stays inside the node's detection area.

In [21] authors proposed about Naive and Prediction based Sequential patterns (PTSP). The naive scheme is one of the basic object-tracking technique, where all the sensors in the OTSN are kept active all the time, where it detects each moving object in the network and report to the base station every T ms. The proposed PTSP is based on two stages: 1) sequential pattern generation and 2) object tracking and monitoring. In the sequential pattern generation stage, from a huge log of data collected from the sensor network and aggregated at the sink in a database the prediction model is built. It produces the inherited behavioural patterns of object movement in the monitored area. From these data, the sequential patterns are generated and deployed by the sink to the sensor nodes in the network. Therefore the future movements of a moving object is predicted by the sensor nodes, if the object is in their detection area. The second stage consists of the actual tracking process which has two parts: 1) activation mechanism, and 2) missing object recovery mechanism. The first method entail the use of the sequential patterns to predict which node(s) should be activated to continually keep tracking of the moving object, and the second method is used to find missing objects in case the activated node is not able to locate an object in its detection area.

In Prediction based approach there exists an algorithm called DPR (Dual Prediction-based Reporting). It mainly focuses on reducing the energy by keeping most of the nodes in sleep mode. Here the next location of target is calculated at both sensor nodes and sink. When the difference between real location and predicted location is acceptable, no update message send to sink and therefore the number of packets transmitted decrease.

3. Prediction based clustering algorithm

Cluster analysis or clustering is the task of assigning a set of objects into groups (called clusters) so that the objects in the same cluster are more similar (in some sense or another) to each other than to those in other clusters. Cluster-based architecture are more advantageous which provides scalable and resource efficient solutions. In proposed algorithm we present an energy efficient prediction-based method in a clustered network. Our algorithm provides the network with grids. A sensor node is selected as Cluster-head in each grid that has higher energy and longer range than other nodes. Only CHs are active and other nodes of the network are in sleep mode at the first of network deployment. The first CH that detects the target becomes active, when a target enters the network. Then three other sensor nodes that are member of the CH are selected by the CH for tracking. The target is being sensed by the nodes and current target location is calculated from this information.

Localization of target is the main idea of our algorithm, via tri-lateration algorithm. This is carried out locally in sensor nodes not in CHs. The details are as follows

Define:

thr: user-defined error threshold
 PL: Predicted next location of target.
 CL: Current location of target.

Algorithm:

1. The next location of the target is predicted by the active CH via prediction mechanisms.
2. If (PL is placed in current cluster) then
 - Active CH:
 - a. Three nodes that are near to the PL are selected.
 - b. Selects one of them as leader node.
 - c. Sends a message to leader node which contains PL.
 - d. Sends messages to two other selected nodes. In these messages inserts leader node ID.
 - Each of selected nodes (except leader node).
 - a. Calculates its distance from the target.
 - b. Sends a message that contains calculated distance and its coordinates to the leader node.
 - Leader node:
 - a. Calculates its distance from the target.
 - b. Receives distance messages from two selected nodes.
 - c. Calculates CL using trilateration algorithm.
 - d. If $|CL-PL| > thr$ then
 - Sends a message that contains CL to the active CH.
 - Else
 - Sends a message to the nearest CH to PL and this CH becomes active CH.
3. If (active CH receives a message from leader node) then
 - Sends CL that located inside this message to the sink.
 - Else
 - Sends PL to the sink.

As see in the above protocol algorithm, localization of target location is carried out locally by leader sensor nodes not centrally by CHs. This method uses two of selected sensor nodes for tracking in each period time. After this their distance is sent from the target to third (leader node) and localization is done by the leader node. Whereas the previous prediction-based target tracking methods consists of three selected sensor nodes, send their distance from the target to CH separately and CH does localization.

Our method helps us to reduce energy consumption of the network as transmission power is directly related to the distance. The reason is that the three nodes are often close to each other for tracking and consumed energy for sending a message between them is lower than consumed energy for sending a message from one of them to CH. Also because when the leader node while sending a message to CH that difference between predicted

location that has received from CH and the calculated current location is more than error threshold defined by user, the number of transmitted packet and direct reduction of energy consumption takes place.

3.1 Prediction mechanism

Prediction-based algorithms are mainly used to predict the next location of the target. A linear prediction method which is a prediction mechanism is introduced in our algorithm. This mechanism predicts the next location of target with the current and previous location of target. So the target speed is calculated as

$$v = \frac{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}}{t_i - t_{i-1}} \quad (1)$$

While the direction is given by

$$\theta = \cos^{-1} \frac{x_i - x_{i-1}}{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}} \quad (2)$$

The predicted location of target is calculated after a given time t with below relations, based on the above information.

$$x_{i+1} = x_i + vt \cos \theta \quad (3)$$

$$y_{i+1} = y_i + vt \sin \theta \quad (4)$$

After calculation (x_{i+1}, y_{i+1}) , if the current cluster consists of this location, then the CH that is active selects three sensor nodes to the nearest predicted location, and wakes them up by sending a message for target tracking in the next interval. Otherwise if the current cluster does not consist the next location of target, the nearest CH is selected by the active CH to that location as next active CH and with sending a message informs it from arriving the target and the tracking task is given to the new active CH.

3.2 Trilateration algorithm

As soon as the distance message is received by the leader node from two other selected sensor nodes, the current location of the target is calculated. We use trilateration algorithm for localization, for this purpose. This algorithm helps us to form the below relation for three nodes and from solving three formed relations, the target's coordinate is obtained.

$$(x_i - x_t)^2 + (y_i - y_t)^2 = d_i^2, (i = 1, 2, 3) \quad (5)$$

In this relation, (x_i, y_i) are coordinate of node i , (x_t, y_t) are coordinate of target and d_i is distance of node i from the target

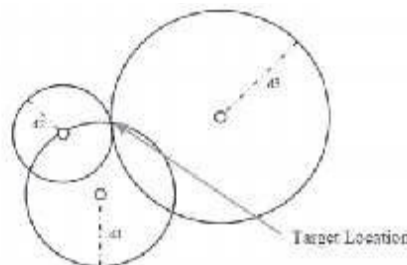


Fig.1. Trilateration Algorithm

4. Experimental set up and results

In this section, using network simulator tool, we evaluate performance of proposed algorithm. Our simulation has done in network simulator 2. We compare our proposed method with previous prediction-based algorithms. We use three below parameters for evaluate performance of our algorithm with others:

- The number of dead nodes of network over time.
- Network lifetime: The time that the first node of network dies.
- Average Energy Consumption (AEC) of nodes.

4.1 Environmental setup

A network is considered in the simulation model where the variation of the nodes is made between 120 and 200. Eight CHs are included in this network. Uniform distribution is done for CHs and random distribution for other nodes between $(x = 0, y = 100)$ and $(x = 400, y = 300)$ with sink at location $(x = 50, y = 50)$. The initial energy of each normal node is 0.5J. The target speed is varied between 0 and 3m/sec and target trajectory is chosen randomly. Tracking period and user-defined threshold is set to 1sec and 1 meter. Simulation time is 30 minutes.

In this model, the energy consumed to transmit a packet to a distance $d(E_t)$ and energy consumed for receiving a packet (E_r) are as follows:

$$E_t = C + K * d^2 \quad (7)$$

$$E_r = C \quad (8)$$

We used $C = 60$ nJ/bit and $K = 20$ pJ/bit, in our Simulation.

4.2 Simulation and Performance Evaluation

The performance of our algorithm is evaluated in this section. Figure 2 shows the number of dead nodes of network during simulation in one random scenario. In this scenario the number of nodes in this network is considered 200. When compared with other prediction-based methods, the proposed algorithm consists of less number of dead nodes as expected. This is because in each tracking period of time in proposed algorithm, only one node is assigned as leader node that does some tasks of CH (receiving distance message from two other selected nodes and calculating current target location).

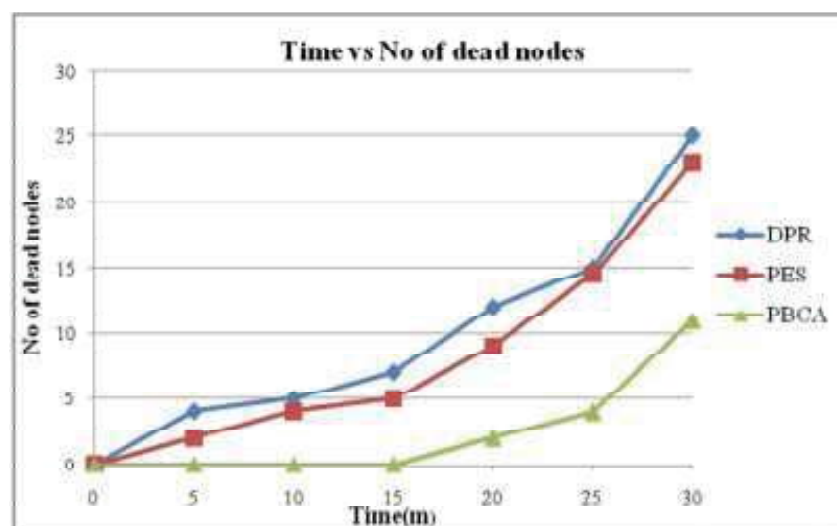


Fig.2. Number of dead nodes

Whereas for tracking, the distance between three selected nodes is often very lower than their distance from CH, by our method the lifetime of nodes and network lifetime will be increased and also energy consumption of network will be decreased. Figure 3 shows AEC of nodes versus time in compared algorithms.

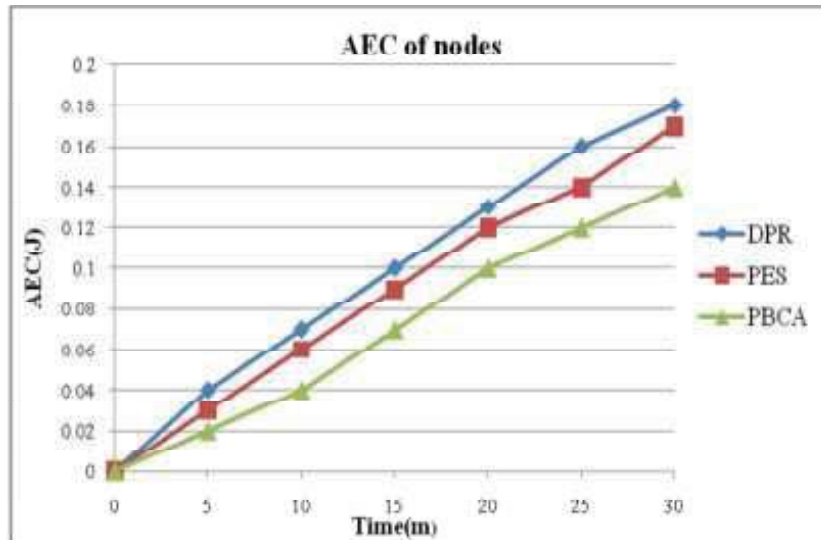


Fig.3. Average Energy Consumption (AEC) of nodes

For more comparison results, our simulation can be made to run in many different scenarios, with different node number, and the network's lifetime is calculated. Figure 4 (a) shows average lifetime of network versus node number. Since the average network lifetime of our proposed algorithm is more, when compared with previous prediction-based algorithm, time increases as the number of nodes in the network increase. Also the average number of dead nodes for these scenarios in 30 mins of simulation is depicted in fig 4 (b).

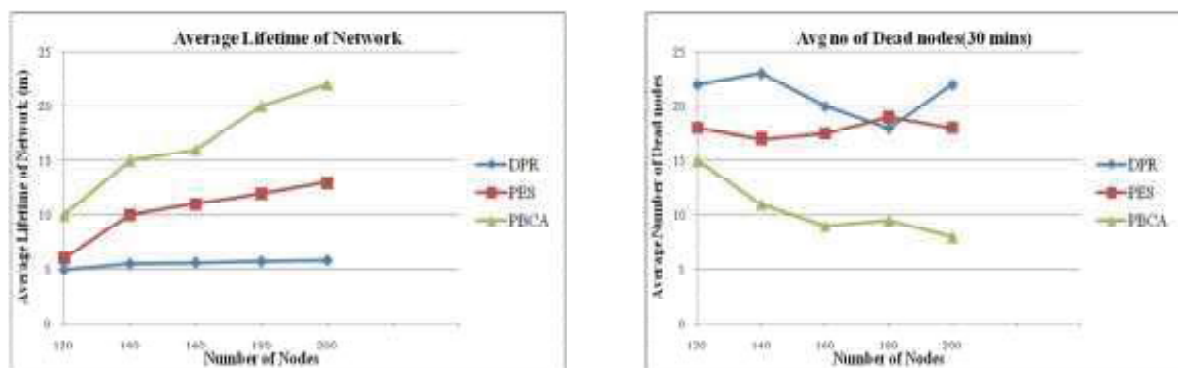


Fig.4. (a) Average Lifetime of Network; (b) Average Number of Dead Nodes (in 30 mins)

5. Conclusion

An energy efficient prediction- based clustering algorithm is proposed in this paper, for target tracking in WSNs. Localization of moving target is done locally in proposed algorithm and not centrally by CHs. This method helps in the reduction of the transmission distance in sending localization packets and therefore consumed energy will be decreased for sending these packets. Moreover our algorithm, by using a prediction mechanism, only when the difference between the calculated current location with sensor nodes and predicted location with CH has been more than a predefined threshold, sends the calculated current location to CH and there takes place a reduction in the number of transmitted packet between sensor nodes and CHs. Simulation results show that the average energy consumption by sensor nodes are decreased, and therefore the life time of the network is increased when compared with other prediction based algorithms.

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